

Solvency Field Theory V13.12

THE MAINTAINED ELECTRON

A parameterized stable recurrent null closure with finite support, spin-1/2 framing, charged dressing, and one-state physical readouts

Public-facing V13.12 draft v0.2 built from the C17G19 full freeze of the SFT Electron V1.

This document presents the frozen maintained-electron model in structural order rather than discovery order. Section 1 shows the electron first: what SFT says the object is, how its parts work together, and what the mathematics says at the current V1 boundary. The remaining sections then justify that picture, expose its parameters and interfaces, and preserve the negative controls that prevent overclaiming.

Freeze status

The numerical reference state is frozen. This public document is a presentation draft, not a reopening of the electron solve. Any future change to the maintained state must fork a new electron version rather than silently overwrite C17G19.

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External review: Claude / Fable

Independent human-AI research collaboration | August 2026

Model status: PARAMETERIZED STABLE MAINTAINED SFT ELECTRON V1

Frozen-state SHA-256: ba57e94177dae9f6ff2b734ae25caa05105b7aa7d9636e39403022843f1dd6dd

Abstract

In SFT, an electron is modeled as locally null propagation organized into one self-maintaining finite recurrent structure. V13.12 presents the public formulation of the maintained Electron V1 frozen at C17G19. The model does not treat the electron as a point object moving inside a separate container or as a photon traveling on a pre-existing circular track. Instead, its local null propagation, finite transverse source, distinct quadratic settlement margin, self-consistent metric response, round scalar support, half-twisted spin/framing sector, curved electromagnetic dressing, and recurrence are treated as aspects of the same maintained state.

The frozen numerical representative is parameterized rather than absolute. Mature maintenance stabilizes a supplied support normalization Γ_{star} but does not select its formation value. The connected scalar support is round; the native rest branch is Gordon/Pauli with ordinary charge-orbital $L=0$, spin $S=\hbar/2$, and tree-level $g=2$; the compact source is unit normalized to numerical precision; the complete recurrent route has vanishing net spatial momentum; a post-closure cycle supplies a dimensionless clock; and the reaction-completed V1 perturbation map is elliptic and bounded through hostile and long-cycle tests. All of these readouts are taken from the same frozen state without observable-specific retuning.

The charged representative does contain a non-negligible electromagnetic Hilbert dressing whose ancestry used $\alpha=1/137$ as an external correspondence normalization. A frozen-state ablation finds no common residual selector at that value, so V13.12 exposes electromagnetic stiffness/absolute charge normalization as an interface input rather than claiming α as an electron-derived constant. Whole-space Coulomb and ADM exteriors, absolute SI rest scale, intrinsic α , the anomalous magnetic moment, formation and destruction, exchange statistics, and real interactions remain separate downstream programs.

Strongest honest claim

SFT Electron V1 supplies one stable finite recurrent charged spin-1/2 maintained structure at parameterized support scale, with dimensionless post-closure clock/rest-frame and leading Pauli readouts on one frozen state; absolute support scale, electromagnetic coupling normalization, SI rest scale, global exteriors, formation, and interactions remain separate open programs.

Collaboration disclosure. This manuscript was developed through a human-AI research workflow. Shaun Fosmark directed the SFT program, theory decisions, artifact selection, and publication boundary. GPT-5.6 Sol contributed derivation, numerical-audit design, synthesis, and manuscript drafting. Claude / Fable provided adversarial external review. Numerical claims were retained only where preserved by the frozen artifact chain and, where applicable, survived independent holdout or ablation checks. This disclosure does not substitute for independent peer review.

Contents

1. The SFT Electron: What It Is and How It Works	11. Parameter, Interface, and Dependency Ledger
2. The Maintained-Electron Problem	12. Negative Controls and Superseded Pictures
3. The Exact Finite Source	13. Claim Boundary
4. Recurrent Geometry and Closure Gravity	14. Discussion: Particle as Recurrent Structure
5. Finite Support Without a Hard Shell	15. Interactions and the Next Falsification Program
6. Spin-1/2 Framing and the Pauli/Gordon Branch	Conclusion
7. Curved Electromagnetism and the Alpha Interface	References and Companion Documents
8. The Frozen Common Electron	Appendix A. Numerical Certificate
9. Stability of the Maintained State	Appendix B. Internal Provenance Map
10. One-State Physical Readouts	Appendix C. Reproduction and Freeze Rule

1. The SFT Electron: What It Is and How It Works

The electron, in one sentence

In SFT V13.12, an electron is a finite, persistent recurrent null structure whose local causal propagation, transverse support, settlement response, metric, electromagnetic dressing, spin/framing, and recurrence are different descriptions of one maintained physical state.

This section intentionally begins with the finished object. The aim is that a reader can understand what SFT is proposing before being asked to follow the derivation, numerical gates, or claim firewalls. The rest of the paper then asks whether each part of this description is actually supported by the frozen V1 model.

SFT V13.12 electron architecture

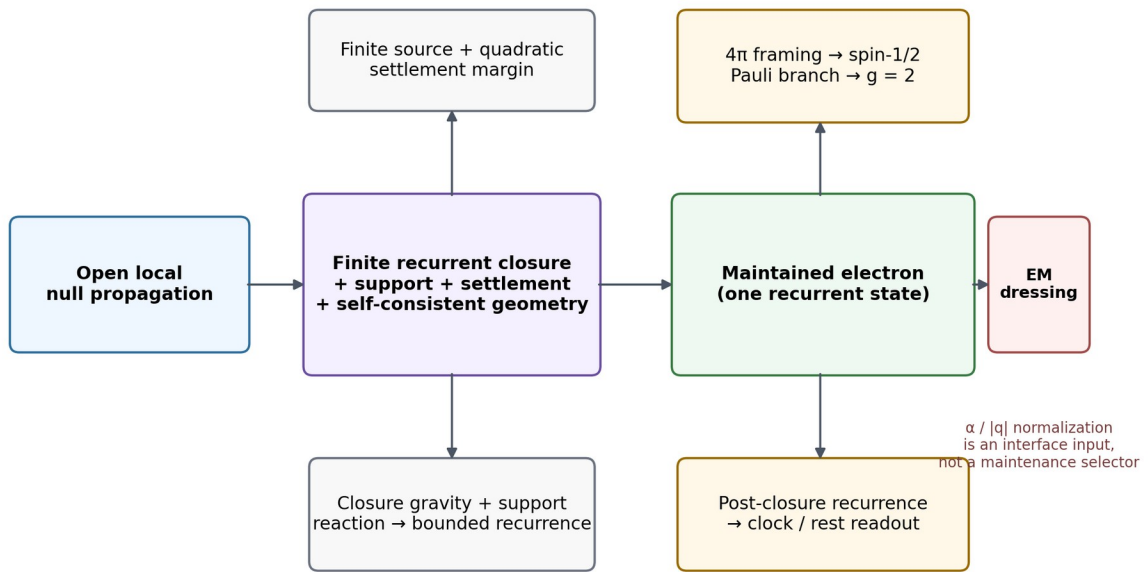


Figure 1. Structural map of the maintained SFT electron. The arrows indicate dependency and readout order, not a chronological construction history.

1.1 One object, not a carrier inside a container

In one sentence: an electron in SFT is locally null propagation organized into a self-maintaining finite recurrence. A photon is open null propagation. A massive particle is null propagation organized into persistent closure, and the electron is the charged spin-1/2 branch of that closed class. The local tangent remains null, but the complete route closes; the integrated spatial momentum cancels; the finite support and geometry recur; and only after that recurrence exists can an observer identify a period and call it a clock.

open null propagation -> finite recurrent closure -> particle -> clock readout

The route is therefore not a track laid down in advance, and the support is not a shell surrounding a smaller object. A useful state notation is

$$S_e = \{ k^\mu, f_{\text{perp}}, q_{\text{perp}}, g_{mn}, A_\mu, D, H \}$$

where k is the locally null current/tangent, f_{perp} is the finite carrier profile, q_{perp} is the distinct quadratic settlement margin, g_{mn} is the common metric response, A_{mu} is the electromagnetic connection/field sector, D is the transported director or framing state, and H denotes the irreversible history/recurrence bookkeeping. These are not separate particles or energy stocks.

1.2 The common-root statement

The maintained-electron question is whether the complete state can be stationary in the appropriate material description while satisfying its common equations and recurrence conditions together. Schematically,

$$\delta S[S_e] / \delta S_e(\theta) = 0$$

subject to local nullity, finite support, source conservation, Einstein/source compatibility, Maxwell compatibility, recurrence, framing return, and stability. θ here is only a coordinate around the closed structure. It is not an upstream clock.

Scope of the word solution

The C17G19 object is a frozen numerical V1 state at a practical Pareto floor, not a proof of an exact analytic global root. “Electron solution” in this document means the validated parameterized numerical maintained state at the explicitly stated V1 scope.

1.3 How local null propagation becomes a rest particle

At every occupied point, the carrier sector is locally null. The direction changes around the recurrent route because the route belongs to the solved geometry. Over one complete closure the directional contributions cancel. A fresh high-resolution route quadrature gives

$$|P_{\text{vec}}| / E = 3.451729874 \times 10^{-15} \sim 0$$

so the recurrent structure possesses a rest-frame spatial momentum readout even though its local causal process is not stopped. This is the central kinematic move of the model: rest is a property of the complete closed structure, not of a locally motionless constituent.

1.4 What gives the electron finite size

The electron carries a finite transverse source rather than a delta-function line. That source is inherited from the nodeless one-write G12U mark and projected onto the transverse screen of the recurrent worldtube. The model keeps two different profiles: a linear/current carrier f_{perp} and a more centrally concentrated quadratic settlement margin q_{perp} . The latter is not obtained by squaring an already-projected carrier profile; it is a separately projected quadratic source object.

$$\begin{aligned} f_3(r) &= (1/4) \sin(\pi r) / r, \quad 0 < r < 1 \\ f_{\text{perp}}(u) &= (1/2) \int_0^1 [\sin(\pi r) / \sqrt{r^2 - u^2}] dr \\ q_{\text{perp}}(u) &= 2 \int_0^1 [r f_3(r)^2 / \sqrt{r^2 - u^2}] dr \end{aligned}$$

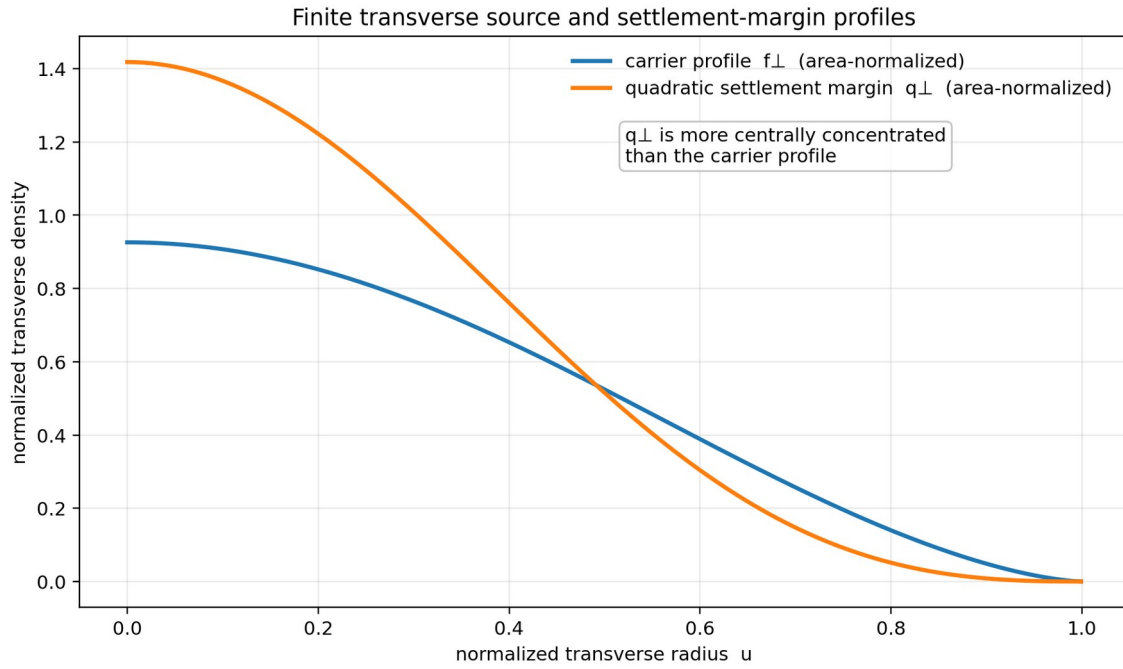


Figure 2. Area-normalized shape comparison of the finite carrier profile and the distinct quadratic settlement margin. The margin is more centrally concentrated.

The connected intrinsic-even branch selects a round scalar support. The equilibrium ellipticity variable is

$$r_{eq} = 0$$

but round scalar support does not imply trivial internal orientation: the spin/framing sector can twist on a circular support.

1.5 What holds the electron together

No hard membrane, conductor, or material wall is inserted. The finite source participates in the metric it follows. The compact local settlement response bends and branch-locks the recurrent structure; SFT reserves the phrase closure gravity for this strong compact microscopic action and uses gravity for the dilute macroscopic exterior readout. The lawful support reaction can carry stress without introducing a second positive energy reservoir: its constraint vanishes on the legal branch even though metric variation can produce nonzero support stress.

Perturbations are then confined by the reaction-completed optical/director map. A static saddle would focus one transverse direction while defocusing the other; the physical co-rotating framing turns this alternating-gradient structure into a bounded recurrent map. The later stability section tests that mechanism rather than assuming it.

1.6 Why a clock and mass readout appear only after closure

A free photon has no intrinsic rest clock in this picture. The first intrinsic tally appears when a finite structure returns to an equivalent state and can be counted cycle by cycle. The frozen null-cycle readout is

$$T_{cycle} c / R_{coord} = 12.07620577648$$

The period is downstream. It was not used to tune the structure that produced it. Only after closure may one invoke the mature action-phase correspondence

$$E T_{\text{cycle}} = h$$

which gives the dimensionless post-closure scale relation

$$E R_{\text{coord}} / (\hbar c) = 0.52029465409$$

The coefficient 0.52029465409 is about 4.1% above the idealized 1/2 closure value. V13.12 does not promote that difference as a new physical correction: R_{coord} is a reference chart scale, and C17G18 explicitly classifies route-scale readouts as chart/scale conditional. The invariant content claimed here is the post-closure identity $E T_{\text{cycle}} = h$ together with near-zero total spatial momentum; a coordinate-invariant radius theorem remains open.

An absolute electron mass in kilograms is therefore not predicted by the maintained V1 alone, because the physical value of the coordinate/support scale is not selected here. One later rest-scale anchor could convert the dimensionless state to SI units, but that anchor would be an input, not a prediction. For context, the Standard Model likewise does not derive the charged-lepton masses from gauge structure alone: the electron, muon, and tau masses are set by independent Yukawa couplings once the Higgs scale is specified [E1]. V13.12 therefore treats its one-anchor economy as a testable compression, not as permission to hide the remaining dimensional input.

Order-of-operations firewall

Closure creates the first ruler/clock. The ruler/clock is not allowed upstream to normalize or select the dynamics that create the closure.

1.7 Why the electron has spin-1/2 and tree-level $g = 2$

The scalar cross-section is round while the transported director/framing state performs a half-twist over the spatial loop and returns on the oriented 4π cycle. The native rest-ground branch is not an ordinary classical charged loop that already carries the full spin. Instead, the Gordon/Pauli partition selects

$$L_{\text{ordinary charge}} = 0, \quad S = \hbar / 2$$

and the first-order Pauli/Clifford operator identity gives

$$(\sigma \cdot \pi)^2 = \pi^2 - q \hbar \sigma \cdot B$$

$$g_{\text{tree}} = 2$$

Spin/framing is not scalar ellipticity

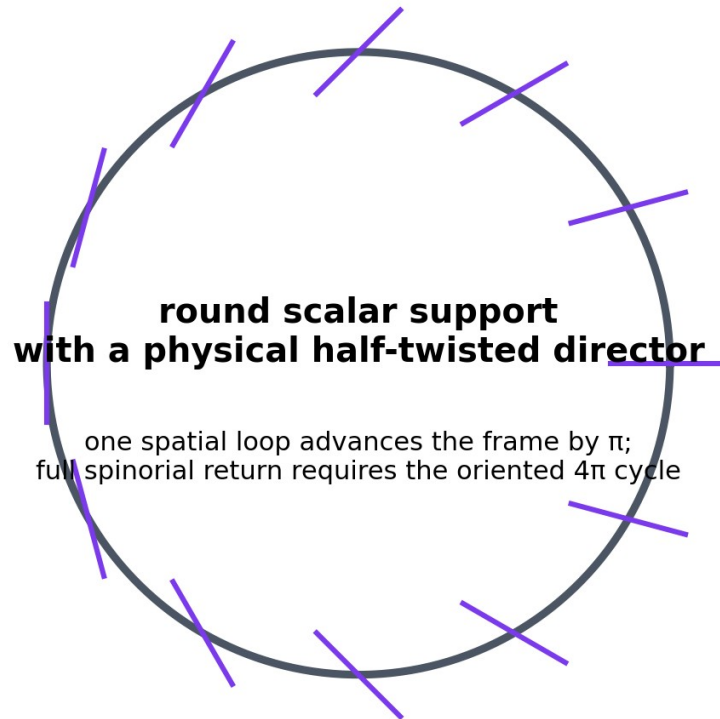


Figure 3. Conceptual framing schematic. The scalar support can be round while a transported internal frame carries a half-twist and requires the oriented 4π cycle for full spinorial return.

This rejects the older literal picture “a $g=1$ charge loop carries $\hbar/2$ and an equal second current is then added.” That decomposition double-counts angular momentum on the modern state.

1.8 What charge means, and where alpha does not belong

The maintained state carries a signed framing/winding branch $n_g = \pm 1$ and a unit-normalized compact source profile. At the dimensionless branch level this supplies charge sign and the Pauli magnetic structure. It does not fix the absolute stiffness of the electromagnetic field. The current charged representative used $\alpha_{EM} = 1/137$ as an external correspondence normalization in its Maxwell dressing ancestry.

charge topology / sign $\neq$ absolute electromagnetic stiffness

C17G18 held the frozen state fixed and rescaled only the electromagnetic stress. The physical ancestry value is not a common optimum of the normalized, raw, and worst-point residuals. The dressing is finite and physically relevant, but its numerical strength is not selected by the maintained electron root. V13.12 therefore exposes alpha and absolute $|q|$ as interface normalization debts rather than hiding them inside the state.

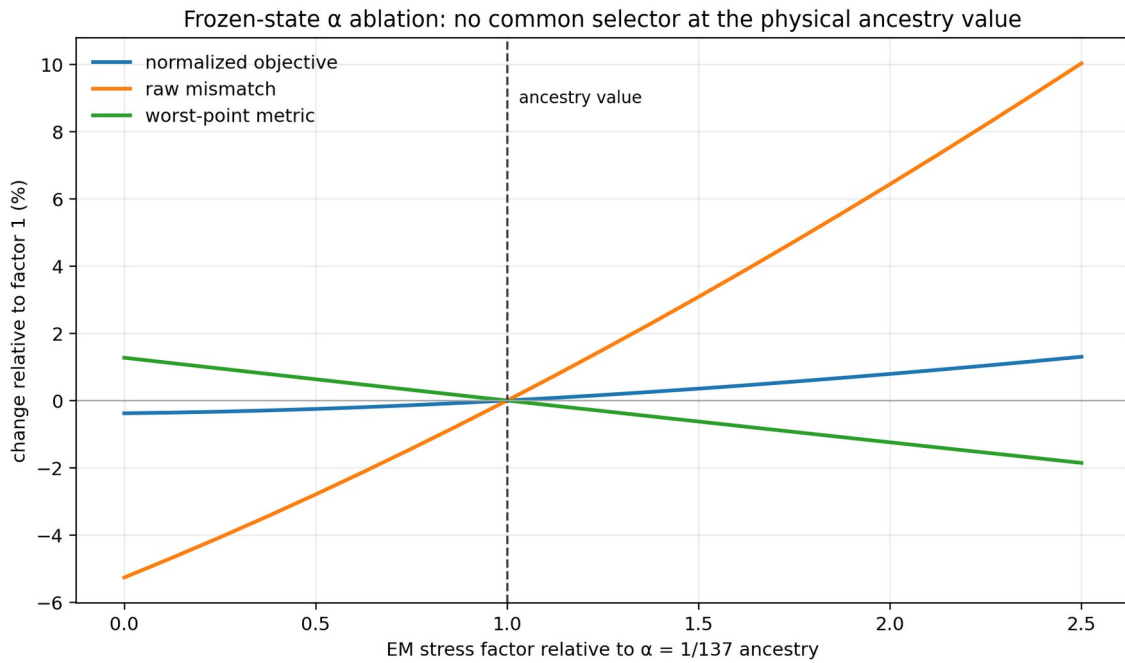


Figure 4. Frozen-state alpha ablation. Different residual measures prefer different directions; there is no common residual selector at the ancestry value $\alpha = 1/137$.

1.9 Why the maintained electron does not radiate away at this scope

The frozen maintained source is stationary in material worldtube coordinates. Its source and field profiles depend on route and transverse position rather than coordinate time. Thus the maintained state is not modeled as a point charge repeatedly accelerating around an externally prescribed loop. At finite-domain maintained-source scope the radiative multipole time derivatives vanish. A complete whole-space outgoing/Sommerfeld theorem remains open and is not silently inferred from the finite correction domain.

1.10 Why the maintained state is stable

The physical perturbation object is the reaction-completed recurrent support map. Its frozen one-cycle Floquet multipliers lie on the unit circle to numerical precision, a positive quadratic invariant exists, and direct powers remain bounded through 100,000 cycles in the invariant norm. Hostile C3-symmetric profile perturbations remain elliptic, while an intentionally amplified saddle crosses the predicted instability boundary and becomes hyperbolic. The control therefore resolves both stability and instability rather than simply returning “stable” for every input.

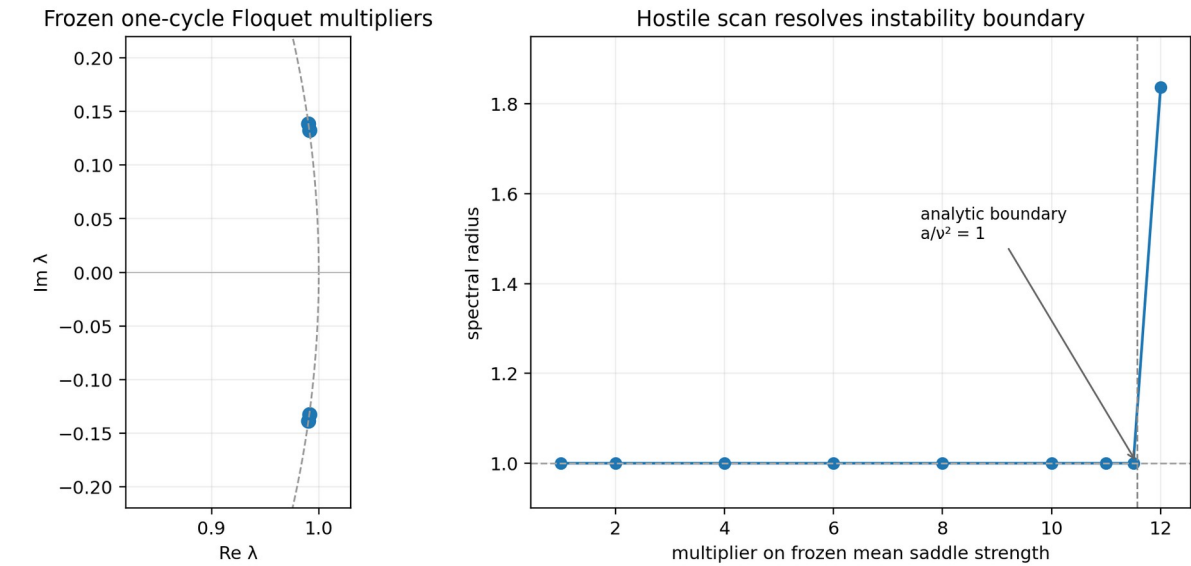


Figure 5. Left: frozen one-cycle multipliers on the unit circle. Right: hostile saddle amplification remains elliptic until the analytic $a/\nu^2 = 1$ boundary is crossed.

1.11 The electron at a glance: what is derived, supplied, and still open

Object	Status	Meaning
Finite recurrent source architecture	PASS	Exact finite carrier + distinct settlement margin; one maintained structure.
Scalar support shape	PASS	Connected branch selects $r_{eq} = 0$.
Spin / leading magnetic branch	PASS	Pauli/Gordon rest branch; $S = \hbar/2$, tree-level $g = 2$.
Maintained V1 stability	PASS	Elliptic reaction-completed recurrence; hostile boundary resolved.
Post-closure clock / rest frame	PASS	Single recurrence period; $ P /E \sim 3.45e-15$.
Local closure gravity	PASS	Nontrivial compact worldtube metric response.
Gamma_star support normalization	FREE PARAMETER	Formation/support input; mature maintenance does not select it.
alpha_EM / absolute $ q $	EXTERNAL / OPEN	Used for charged dressing; not derived or selected by Electron V1.
Absolute SI rest scale	OPEN / OPTIONAL ANCHOR	One downstream dimensional anchor can convert the relational state.
Whole-space Coulomb / ADM exteriors	OPEN	Require explicit asymptotic matching.
Formation, interactions, statistics	OPEN	Separate downstream programs; not used to rescue maintenance.

Internal provenance: C17G15.1-C17G19 frozen stack [P1-P5].

2. The Maintained-Electron Problem

Section 1 described the finished V1 object. We now state the narrower problem that V13.12 actually solved. The question is not “derive every measured property of the electron from zero inputs.” It is: given an already-created electron, can SFT maintain one finite charged spin-1/2 recurrent structure without a hard shell, without an upstream Compton ruler, without duplicated source stocks, and without retuning a different state for each observable?

Maintenance vs formation

Formation is deliberately downstream of this paper’s core claim. A formation theory may eventually select Gamma_star or determine absolute scale. It is not allowed to rescue a maintained structure that fails on its own.

2.1 Foundational constraints

- Local propagation remains causal/null in the carrier sector.
- The particle is the complete recurrent structure, not a photon plus a separate container.
- No-overwrite history may grow, but the active maintained source and geometry recur rather than accumulate unboundedly.
- A clock is a post-closure readout and may not be used upstream to select the closure.
- The source, support, metric, electromagnetic field, framing, and recurrence must coexist in one state.
- No hard shell, conductor, extra membrane, or independent positive support-energy stock may be introduced merely to confine the carrier.
- Observable-specific retuning is forbidden in the final readout audit.
- Open exterior or interaction programs may remain open only if they are not needed to rescue the isolated maintained state.

2.2 Why discovery order is not presentation order

The internal C17 program discovered the electron by a long sequence of gates, negative controls, basis repairs, and dependency corrections. V13.12 does not reproduce that chronology in the main narrative. Instead it presents the minimum structural dependency chain and moves gate-by-gate provenance to Appendix B. This makes the public claim easier to audit: every object has a role, every free parameter is exposed, and every open theorem is separated from what the frozen state actually earns.

3. The Exact Finite Source

3.1 The three-dimensional mark and transverse projection

The finite source starts from the nodeless G12U one-write ground-state mark. In normalized internal radius r ,

$$f_3(r) = (1/4) \sin(\pi r) / r, \quad 0 < r < 1$$

The mature electron does not stack three-dimensional balls around the route. The mark is projected onto the transverse screen of the recurrent worldtube. This produces a finite carrier/current profile f_{perp} and, separately, a quadratic material/settlement margin q_{perp} .

3.2 Carrier and settlement margin are different objects

The distinction is load-bearing. The carrier profile describes where the tangent-null current/source is distributed across the finite cross-section. The settlement margin describes a quadratic density response and is more centrally concentrated. Treating q_{perp} as merely f_{perp} squared after projection changes the physical object and is not the frozen construction.

$$2 \pi \int_0^1 u f_{\text{perp}}(u) du = 1$$

$$2 \pi \int_0^1 u q_{\text{perp}}(u) du = \pi / 8$$

A fresh C17G17 reintegration of the compact source gives 0.9999999864 for the unit normalization. This is the source normalization used by the frozen readout ledger, not an independent charge-magnitude prediction.

3.3 Curved transport of the material margin

In the curved worldtube, fixed material measure μ and physical volume Jacobian J_3 imply $n = \mu/J_3$. Because the retained G12U margin energy/pressure is quadratic in n , its local density scales as J_3^{-2} . C17G14 explicitly retested this identity in the solved spatial metric rather than patching the margin profile when mixed-sector residuals appeared.

$$n = \mu / J_3, \quad \text{epsilon}_{\text{margin}} = p_{\text{margin}} = (K_w/2) n^2 \text{ proportional to } J_3^{-2}$$

Internal provenance: global-periodic C17G14 control and C17G15 structure freeze [P1].

4. Recurrent Geometry and Closure Gravity

4.1 The route belongs to the solution

The frozen ontology rejects a pre-drawn racetrack. The occupied route must be null and geodesically compatible in the same metric sourced by the complete maintained state. This is why a free test-geodesic control can fail to return while the reaction-completed electron remains stable: the physical particle is the coupled structure, not the test trajectory after the rest of the structure has been removed.

4.2 Closure gravity versus macroscopic gravity

Closure gravity names the strong compact microscopic settlement/branch-locking response that turns and maintains the recurrent carrier. Gravity is reserved for the dilute macroscopic exterior readout. The local frozen metric is nontrivial, with representative lapse values spanning roughly 0.856 to 1.023 over the loaded material disk. This local geometry is real inside the model, but the finite correction-domain basis is explicitly matched back to the reference tube at its outer edge. It therefore does not by itself constitute a solved asymptotic $1/r$ gravitational exterior or ADM mass.

No exterior extrapolation by hand

The strong worldtube basis is a finite correction-domain object. Whole-space gravity must be earned by an explicit asymptotic matching theorem or solve; the internal metric may not simply be continued outward and called an ADM exterior.

4.3 One source, multiple readouts

The same maintained source underlies local closure geometry, clock/rest readouts, spin bookkeeping, and the curved electromagnetic dressing. C17G18 found no need for a second gravitational mass stock, a second spin energy, or a second electromagnetic source. This one-count rule is essential to the claim that V13.12 describes one electron rather than a collection of separately normalized submodels.

5. Finite Support Without a Hard Shell

5.1 Round scalar support

The connected intrinsic-even branch selects a round scalar support, $r_{eq}=0$. A twisting internal frame is still permitted and, in fact, is part of the spin branch. Scalar ellipticity and spin/framing are independent pieces of the state.

5.2 The support reaction is not a membrane

The support constraint is zero on the legal branch, so its multiplier carries zero independent on-shell action and performs zero virtual work along admissible motions. Its metric variation can nevertheless contribute support stress. In this sense the reaction acts as a brace of the admissible recurrent domain without becoming a new material wall, conductor, or second positive energy reservoir.

5.3 Confinement as a dynamical property

The stability calculation shows that the maintained transverse map is elliptic rather than hyperbolic. This is the practical confinement statement of V1. The optical/director response keeps small perturbations bounded around the occupied support. Confinement is therefore tested as a property of the recurrent map, not asserted because a finite support radius was drawn into the coordinates.

6. Spin-1/2 Framing and the Pauli/Gordon Branch

6.1 Half-twisted framing on a round tube

The director D transports a physical polarization/frame orientation around the route. The signed branch $n_g = +/-1$ reverses the chirality of that framing while leaving scalar mass-like readouts conjugate-equal. A one-loop half-twist and an oriented 4π return give the state its spinorial framing structure without requiring an elliptically deformed scalar cross-section.

6.2 Why the classical loop picture is rejected

The frozen route can be used as a hostile historical control. If one combines its geometric orbital integral with the post-closure $ET=\hbar$ correspondence, the literal orbital reading is $J/\hbar$ approximately 0.5906 rather than $1/2$. V13.12 retains that mismatch as evidence against resurrecting the old classical-loop ontology. The modern native branch instead uses the Gordon/Pauli partition.

$$\begin{aligned} \mu &= (q/2m) (L + 2S), \quad J = L + S \\ J &= \hbar/2, \quad g = 2 \rightarrow L = 0, S = \hbar/2 \end{aligned}$$

6.3 First-order coupling and tree-level magnetic response

A fresh finite-operator retest closes the Pauli/Clifford identity at approximately $1.22\text{e-}16$ bulk relative error, with the covariant-momentum commutator controlled at approximately $8.22\text{e-}15$. The underlying first-order spinor identity is standard Dirac/Pauli structure [E3]; the SFT claim is narrower: its maintained framing/leading-edge construction lands on that branch without appending a second magnetic current. At this leading operator scope the magnetic response is

$$\mu / [q \hbar / (2m)] = 1$$

$$g_{\text{tree}} = 2$$

This does not derive the anomalous magnetic moment. The standard leading QED correction begins at the Schwinger scale $\alpha/(2\pi)$ [E4]; the V13.12 anomaly remains a higher-order charged-interface/dressing problem.

6.4 One-count angular momentum

Maxwell field angular momentum is not added after declaring the matter sector already equal to $\hbar/2$. The correct bookkeeping is a single total ledger in which matter/spin and the electromagnetic field share the same total angular momentum. Likewise, a separate bare Belinfante tensor is not appended as a second source. Belinfante/Hilbert relations are different descriptions of the same physical source, not extra stocks.

Internal provenance: strict Pauli hot-swap C17G9 and curved Maxwell C17G10 [P6-P7].

7. Curved Electromagnetism and the Alpha Interface

7.1 Electromagnetism is solved on the curved maintained state

The charged representative is not built by pasting a flat Coulomb field onto a finished gravitational geometry. C17G10 solves the stationary sourced Maxwell system through the curved Plebanski constitutive map on each candidate state, enforcing Gauss, no magnetic monopole, curl-E, and Ampere constraints together. At the frozen-lineage base, Gauss/div-B defects are of order $1\text{e-}8$ or better and curl equations close near $1\text{e-}17$. Curvature changes the finite-domain electromagnetic energy by roughly 1.7-3.9% across the directly computed support family.

7.2 Alpha ancestry is exposed rather than hidden

The charged-root lineage used the rounded empirical low-energy correspondence $\alpha_{\text{EM}}=1/137$ to set the absolute amplitude of the electromagnetic Hilbert dressing; the modern CODATA value is $\alpha^{-1} = 137.035999\dots$ [E2]. That ancestry is now part of the public parameter ledger. The strongest possible claim “Gamma_star is the only non-unit continuous number in the numerical representative” is therefore not made.

At the frozen state, sampled electromagnetic stress is about 4.38% of the non-EM source/geometry scale on average and reaches about 11.19% at the largest sampled point. The contribution is secondary but not negligible. The question is not whether electromagnetism matters; it is whether the physical alpha value is selected by the intrinsic maintained equilibrium.

7.3 Frozen-state ablation: no intrinsic 1/137 selector

C17G18 changes no state coefficient. It holds the geometry and source state fixed and rescales only the electromagnetic stress amplitude. On exact train, blind, legacy, and broader frozen panels, turning the dressing off or doubling it produces mixed-sign tradeoffs. On the broad sweep, the raw mismatch and normalized objective do not have a common minimum at factor 1. The physical ancestry value is therefore not an intrinsic residual selector of the frozen maintained structure.

Alpha ruling

alpha_EM is a non-negligible external electromagnetic dressing/interface normalization used by the current charged representative. Electron V1 does not derive or select its numerical value.

This ruling is compatible with, but does not depend on, the earlier standalone SFT Alpha-137 program, which independently left the numerical coupling open and identified a separate microscopic stiffness/accessibility target. That earlier result is companion context, not a rescue input for V13.12 [C1].

8. The Frozen Common Electron

8.1 Why the static structure was frozen

C17G15 removed the final named load-bearing inner-core pin using already-owned low-order geometry rather than new source physics. A fresh 210-point invariant-core holdout verified the improvement. Subsequent lawful same-physics directions continued to improve averages while worsening a previously unseen local invariant-core extreme. Even a 5% step along the leading direction lowered train and mean-core errors but worsened the unseen worst point by 0.211%. The state was therefore typed as a practical Pareto floor for the present V1 representation rather than endlessly polished.

What “frozen” means

The isolated maintained-electron structure is good enough for V1. The freeze does not claim every local numerical residual is exactly zero. Remaining small static mismatch is representation/polish debt unless a downstream audit shows it to be load-bearing.

8.2 Canonical numerical representative

The canonical frozen state is SFT_ELECTRON_V1_STRUCTURE_GOOD_ENOUGH_FROZEN.npz with SHA-256

ba57e94177dae9f6ff2b734ae25caa05105b7aa7d9636e39403022843f1dd6dd

Its support normalization and chart-level support ratio are

$$\text{Gamma_star} = 0.18562349764677286$$

$$s = 0.24965573857744025$$

Gamma_star is not promoted as a predicted electron radius or absolute scale. It is the supplied formation/support normalization of the frozen representative.

8.3 Inherited route labels are scaffolding, not new observables

The numerical tube chart retains historical reference labels $R_{\text{route}}=0.6680425$ and $Z_{\text{route}}=0.75$. C17G14 explicitly released them as live global coordinates while holding Γ_{star} fixed. The equations do see this freedom, but relative to the identical accepted metric move with R/Z frozen, the incremental raw training and held-out improvement was only about 0.04%, with essentially unchanged geodesic closure. The shifted values were deliberately not promoted. V13.12 therefore classifies R/Z as reference embedding scaffolding, not physical electron shape predictions.

9. Stability of the Maintained State

9.1 Physical stability object

Because the electron is the recurrent structure itself, the load-bearing stability map is the reaction-completed recurrent optical/support map, not an unconstrained test photon moving after the support reaction has been switched off. The frozen support gives $a/\nu^2 = 0.0864525425$, well inside the strict elliptic interval $0 < a/\nu^2 < 1$. The analytic boundary lies a factor 11.567 above the frozen mean saddle strength.

9.2 One-cycle Floquet certificate

Fresh monodromy yields multiplier pairs $0.99117056 \pm 0.13259307 i$ and $0.99037325 \pm 0.13842265 i$. Their moduli are unity to numerical precision, with maximum unit-circle defect $6.48e-13$ and symplectic residual $5.39e-12$. A strictly positive quadratic invariant $M^T G M = G$ exists with residual $1.72e-13$. No physical multiplier is close to +1, removing the simple cycle-locked secular-resonance route in the linear support map.

9.3 Long-cycle and hostile tests

After transforming to the positive invariant norm, direct powers through 100,000 cycles reach a maximum amplitude of 1.0000000463 and maximum invariant-norm drift of $1.25e-7$. One hundred and five fresh hostile C3-symmetric profile perturbations produce zero unstable cycles and zero failures of positive-invariant construction. Multiplying the frozen mean saddle up to factor 11.5 remains elliptic; factor 12 crosses $a/\nu^2=1$ and produces spectral radius 1.83639 . The control therefore finds the expected instability when one is deliberately created.

9.4 Free-geodesic negative control

Removing the maintained support/reaction interpretation and launching an unconstrained null test geodesic from the prescribed centerline misses the one-route return, with reduced transverse return norm approximately 0.79403 . This is not a failure of the maintained electron; it demonstrates that the stable object is not “a photon on a static racetrack.” The support/metric/current response is part of the particle.

Stability scope

PASS: reaction-completed maintained V1 perturbation dynamics. OPEN: an arbitrary fully nonlinear time-dependent Einstein-Maxwell-material stability theorem for all possible spacetime/source perturbations.

10. One-State Physical Readouts

C17G17 and C17G18 deliberately read every claimed observable from the same frozen SHA without state retuning. The point of this section is not that every laboratory-unit constant is derived; it is that the surviving clock, rest-frame, spin, charge, radiation, and local-gravity statements can coexist on one electron without contradictory source accounting.

10.1 Clock and rest-frame chain

$$T_{\text{cycle}} c / R_{\text{coord}} = 12.07620577648$$

$$L_{\text{proper}} / R_{\text{coord}} = 12.06354587887$$

$$L_{\text{proper}} / (c T_{\text{cycle}}) = 0.99895166596$$

The small difference between spatial proper length and cT is carried by the frozen lapse/shift geometry. Spatial path length is therefore not substituted for the null-cycle clock. With the downstream $ET=h$ relation, $E R_{\text{coord}}/(\hbar c)=0.52029465409$ and the closed-route P -vector vanishes to about $3.45e-15$ relative to E .

10.2 Charge and magnetic moment

The compact source is unit normalized and the signed framing sector is retained. At leading Pauli scope, $\mu/[q \hbar/(2m)] = 1$. Absolute q , α , and SI magnetic moment remain open until electromagnetic and rest-scale normalizations are supplied. No third observable-specific magnetic coefficient is introduced.

10.3 Stationary maintained-source nonradiation

The frozen source is stationary in material coordinates: Lie derivatives of T_{mn} and J^μ along the stationary time direction vanish in the maintained representation. Accordingly, ordinary radiative multipole time derivatives vanish at this finite-domain scope. A whole-space Sommerfeld theorem remains open.

10.4 Local gravity face

The same normalized source that carries the maintained structure produces the nontrivial local closure-gravity metric. No second gravitational mass stock is inserted. A whole-space $1/r$ exterior and ADM mass remain open because they require an explicit asymptotic continuation beyond the matched finite correction domain.

10.5 One-state consistency result

C17G18 result

No surviving readout requires a different frozen electron, an extra positive source stock, or an observable-specific coefficient change. α is exposed as an external EM interface and inherited route R/Z as numerical scaffolding. Under those corrections, there is no blocking internal contradiction.

11. Parameter, Interface, and Dependency Ledger

The model should be judged with its parameters visible. V13.12 therefore separates intrinsic model inputs, derived quantities, discrete branch labels, external interfaces, numerical scaffolding, and downstream open theorems.

Object	Status	Meaning
Gamma_star	INTRINSIC FREE PARAMETER	0.18562349764677286 in frozen representative; formation/support normalization.
r_eq	DERIVED / FIXED	0; connected scalar support is round.
eta_mu / Pauli completion	DERIVED / FIXED	1; leading Pauli/Clifford branch gives $g=2$.
n_g	DISCRETE BRANCH	+/-1 framing/charge-conjugate orientation.
alpha_EM	EXTERNAL INTERFACE	1/137 used in charged ancestry; not derived or selected by V13.12.
absolute q	OPEN INTERFACE	Signed dimensionless source exists; Coulomb normalization not predicted.
rest-scale anchor	OPTIONAL POST-FREEZE INPUT	Can convert dimensionless state to SI; not a prediction.
R_route, Z_route	NUMERICAL SCAFFOLD	Inherited chart labels; not physical electron predictions.
route segmentation	NUMERICAL RESOLUTION	Quadrature partition; physical route must converge under refinement.
old q_h, I_active/i_event	SUPERSEDED	No fundamental six-slot or exponential-memory ontology after continuous-worldtube correction.

11.1 Dependency order

1. Local causal propagation and finite source architecture are defined without using a post-closure clock.
2. The common recurrent structure closes in its support/metric/framing/EM system at supplied Gamma_star.
3. Recurrence then creates the first intrinsic cycle readout.
4. Clock/rest-energy correspondence is applied downstream, not as a selector.
5. Absolute electromagnetic stiffness remains an external charged-interface normalization.
6. Whole-space exteriors and interactions are tested only after the isolated maintained state is frozen.

12. Negative Controls and Superseded Pictures

A public model is easier to understand when it says not only what it uses, but what close alternatives were tested and rejected. These negative controls are not historical clutter; they define the claim boundary.

Object	Status	Meaning
Photon on a static racetrack	REJECTED	Free-geodesic control misses recurrence; the physical object is the complete coupled structure.
Hard shell / conductor / membrane	REJECTED	Finite support is a smooth material/current domain with lawful reaction, not a separate wall.
Repeated laps add mass/curvature	REJECTED	Mature recurrence refreshes the active source; history grows without secularly stacking the maintained energy stock.
Wavelength tightens every lap	REJECTED	Frozen state is periodic, not secularly compressed.

Object	Status	Meaning
Classical $g=1$ loop carries full spin	REJECTED AS NATIVE	Frozen hostile readout gives ~ 0.5906 hbar; native branch is Gordon/Pauli.
$g=1$ loop plus equal second current	REJECTED	Double-counts angular momentum/source.
Bare Belinfante tensor appended to Hilbert source	REJECTED	Would count one physical source twice; hostile appendage worsens/null-energy behavior.
Maintenance selects Gamma_star	REJECTED	Maintenance supplies stiffness around a width but has no absolute formation-scale selector.
$\alpha=1/137$ selected by frozen equilibrium	REJECTED	Frozen ablation has no common residual optimum at factor 1.
Literal six physical write slots / exponential forgetting	SUPERSEDED	Continuous-worldtube refinement demotes slots to numerical partitioning.

13. Claim Boundary

13.1 What V13.12 claims

- A finite recurrent maintained electron architecture exists in the present SFT V1 system at parameterized support scale.
- The frozen representative combines local null propagation, finite transverse source, quadratic settlement margin, round scalar support, compact local closure geometry, spin/framing, curved charged dressing, and recurrence in one state.
- The native rest branch is Gordon/Pauli with ordinary charge-orbital $L=0$, spin $S=\hbar/2$, and tree-level $g=2$ at the stated operator scope.
- The complete recurrent source has vanishing net spatial momentum at maintained-source scope and supplies a downstream dimensionless cycle clock.
- The reaction-completed maintained V1 perturbation map is stable under the tested Floquet, long-cycle, hostile-profile, and instability-boundary controls.
- Clock, spin, source normalization, stationary nonradiation, and local closure-gravity readouts coexist on the same frozen state without observable-specific retuning.

13.2 What V13.12 does not claim

- An exact analytic theorem that all field equations vanish globally at machine zero.
- A first-principles formation selector for Gamma_star or an absolute electron radius.
- A prediction of the electron mass in kilograms without a downstream dimensional anchor.
- A derivation of numerical α or the absolute charge magnitude.
- A derivation of the anomalous magnetic moment.
- A solved whole-space Coulomb/Sommerfeld electromagnetic exterior.
- A solved whole-space $1/r$ gravitational exterior or ADM mass.
- A full canonical = Hilbert/Belinfante = ADM total-angular-momentum theorem.
- Arbitrary nonlinear time-dependent Einstein-Maxwell-material stability.
- A scattering/form-factor theorem explaining why a finite maintained structure is compatible with the experimentally pointlike electron response to momentum-transfer probes; this is OPEN downstream of the interaction program [E1].
- Pair creation/annihilation, two-particle force laws, exchange antisymmetry, or complete indistinguishability statistics.

- A many-particle macroscopic-gravity coarse-graining theorem.

Public wording rule

Do not describe V13.12 as “the electron derived from first principles with no free parameters.” The frozen result is stronger when stated accurately: a parameterized stable maintained electron with a small, explicit interface/open-debt ledger.

Scattering / form-factor firewall

V13.12 does not yet explain why a finite maintained worldtube should present no resolved electron substructure in high-momentum-transfer scattering. The archived idea that a probe couples to the joint interaction/write configuration rather than directly reading the maintained geometric extent is a hypothesis, not a result. The required scattering operator and form factor must be derived and tested against existing compositeness constraints [E1].

14. Discussion: Particle as Recurrent Structure

14.1 The conceptual compression

The model became smaller as the electron program matured. Several quantities once treated as independent machinery - physical event teeth, six-slot memory, a photon moving inside a particle, scalar ellipticity as spin, a hard confinement wall - disappeared when the state was reformulated as one continuous recurrent structure. The remaining architecture is correspondingly harder to describe as a collection of mechanisms bolted together. Local null flow, closure geometry, support, spin framing, electromagnetic dressing, and clock readout are different faces of one state.

14.2 Rest without local stasis

The most distinctive kinematic statement is that a massive rest state need not be built from a locally stationary primitive. Local propagation can remain null while global recurrence cancels translational momentum and supplies a persistent rest frame. In this picture mass is not a stored substance attached to the carrier; it is a response/readout of a maintained closure that has become capable of recurring and being counted.

14.3 Why the theory boundary matters

The open quantities are not cosmetic. Gamma_star controls the formation/support normalization. Alpha controls electromagnetic stiffness/amplitude. The absolute rest anchor controls laboratory units. Whole-space exteriors determine how the finite internal solution joins asymptotic fields. Keeping these objects separate prevents one successful local model from being quietly inflated into a claim about all of particle physics.

14.4 Reconstruction surplus: outputs that were not fitted

Several features were not empirical targets of the final solve. Support tests selected the round scalar branch $r_{eq}=0$; the Gordon/Pauli partition selected $L=0$ over a classical orbital-current picture; and the 4π framing return survived as a kinematic closure property. The hostile orbital-spin control instead returned $J/\hbar$ approximately 0.5906. These unrequested outputs and nearby failed controls are part of the reconstruction evidence.

15. Interactions and the Next Falsification Program

15.1 The interaction hypothesis

The isolated-electron freeze does not derive a microscopic force law between particles. But the final ontology suggests that interactions may require the same conceptual move that repaired the isolated electron: stop treating a physical relation as something transmitted between permanently isolated objects and instead solve the temporarily joint physical state.

Interaction hypothesis - not yet a theorem

Perhaps fundamentally it is not: “electron A sends a force to electron B.” It may be: “the presence of A and B creates a temporarily joint physical configuration whose lawful geometry differs from either isolated solution, and their resulting motion is what we later describe as a force.”

This statement is intentionally preserved as a future-program hypothesis. Generic perturbations used in C17G16 test the stability of one already-existing electron; they are not a model of real scattering, photon exchange, Coulomb attraction/repulsion, or pair production.

15.2 Near-term falsification program

1. **Formation / Gamma-star selector:** derive or constrain the support normalization without importing the measured electron mass or a post-closure ruler upstream.
2. **Electromagnetic stiffness:** derive the absolute coupling/ α in its own field-stiffness or accessibility program and then rerun the charged representative without correspondence normalization.
3. **Whole-space electromagnetic exterior:** match the frozen compact source to a lawful Coulomb/Sommerfeld exterior without retuning the internal state to hit the answer.
4. **Whole-space gravity:** match/coarse-grain the same frozen source into a dilute exterior and test ADM/ $1/r$ behavior.
5. **Two-body joint-state solve:** start with two frozen isolated electrons, construct the joint admissible geometry/source state, and test whether relative motion reproduces the expected charge-sign response without inserting an external force law.
6. **Particle-photon joint-state tests:** replace generic perturbations with relational joint solutions and recover the appropriate low-energy scattering limits.
7. **Electron form factor / apparent pointlikeness:** derive the probe operator and determine why momentum-transfer experiments do or do not resolve the maintained structure. The current archive sketch - that scattering reads the joint interaction/write configuration rather than the isolated geometric extent - is explicitly a hypothesis; compare the result against existing compositeness constraints [E1].
8. **Exchange / indistinguishability:** determine whether two instances of the same root admit the required antisymmetric/composite state structure.
9. **Nonlinear stability extension:** promote the stationary V1 support map to a fully time-dependent Einstein-Maxwell-material evolution and test arbitrary finite perturbations.

15.3 A simple stop rule

Each downstream program should preserve the same discipline used here: if a new observable requires changing the frozen isolated electron only to make that observable work, V13.12 has not been extended - it has been falsified or superseded and a new electron version must be declared. If the frozen state survives and the new program adds only a lawful exterior, interface, or joint configuration, the maintained V1 remains the correct parent.

Conclusion

SFT V13.12 packages the C17G19 frozen Electron V1 into a public structural statement. The electron is modeled as one finite recurrent null structure. Its finite source, quadratic settlement margin, local closure geometry, round support, spin/framing branch, charged Maxwell dressing, and recurrence belong to the same maintained state. That state is stable under the reaction-completed V1 dynamics actually defined by the model, possesses a post-closure clock and rest frame, and supports the leading Pauli/Gordon spin and magnetic readouts without a separate classical orbital-spin stock.

The result is intentionally parameterized. Γ_{star} remains a formation/support free parameter; α and absolute charge magnitude remain electromagnetic interface debts; an absolute SI rest scale requires a downstream anchor; whole-space exteriors, formation, interactions, and exchange statistics remain open. None of those open objects is used to rescue the frozen maintained structure.

V13.12 in one line

A stable maintained electron emerges in SFT as finite recurrent local-null closure - not as a point object on a track - with its clock, rest frame, spin, charged response, and local closure geometry read from one frozen state under an explicit parameter/interface boundary.

References and Companion Documents

External references are used for standard experimental/theoretical context only; the SFT numerical claims are controlled by the frozen internal provenance in Appendix B.

- [E1] S. Navas et al. (Particle Data Group), “Review of Particle Physics,” Phys. Rev. D 110, 030001 (2024), with 2025 update. Used here for Standard Model lepton-mass context and experimental compositeness/scattering constraints.
- [E2] P. J. Mohr, D. B. Newell, B. N. Taylor, and E. Tiesinga, “CODATA recommended values of the fundamental physical constants: 2022,” Rev. Mod. Phys. 97, 025002 (2025).
- [E3] P. A. M. Dirac, “The Quantum Theory of the Electron,” Proc. R. Soc. Lond. A 117, 610-624 (1928), and A 118, 351-361 (1928).
- [E4] J. Schwinger, “On Quantum-Electrodynamics and the Magnetic Moment of the Electron,” Phys. Rev. 73, 416-417 (1948).
- [C1] SFT Alpha-137 Standalone Note (June 2026), companion manuscript. Preserves the separate α /stiffness-accessibility program and explicitly does not claim a numerical derivation of α .
- [C2] SFT IEMM C17G19 Full Freeze Pack (August 2026), internal reproducibility archive for the frozen maintained Electron V1.

Appendix A. Numerical Certificate

The table below collects the frozen values most useful for independent auditing. Values are descriptive of the C17G19 representative at the stated scope and should not be promoted beyond their classification.

Quantity	Frozen value	Classification / use
Frozen SHA-256	ba57e...d6dd	Exact state identity through structure, stability, readout, and consistency gates.
Gamma_star	0.18562349764677286	Intrinsic formation/support free parameter in frozen representative.
support ratio s	0.24965573857744025	Chart-level readout; not an absolute radius prediction.
T_cycle c/R_coord	12.076205776483656	Post-closure dimensionless recurrence period.
L_proper/R_coord	12.063545878868466	Frozen spatial proper route length.
ER_coord/(hbar c)	0.5202946540887051	Conditional downstream ET=h readout; chart/scale conditional, not promoted as an exact 1/2 invariant.
P /E	3.451729874e-15	Rest-frame spatial momentum closure.
source normalization	0.9999999864472883	Fresh transverse source reintegration.
g_tree	2	Fresh Pauli/Clifford operator identity.
stability a/nu^2	0.08645254252562229	Inside elliptic interval $0 < a/\nu^2 < 1$.
instability margin factor	11.5670	Mean saddle multiplier before analytic boundary.
unit-circle defect	6.48e-13	Frozen Floquet multipliers.
positive-invariant residual	1.72e-13	$M^T G M = G$.
100k-cycle max invariant amp	1.0000000463	Long-cycle boundedness test.
hostile profiles unstable	0 / 105	Fresh deterministic C3 perturbation family.
instability control	factor 12 -> rho=1.83639	Correct hyperbolic transition after $a/\nu^2 > 1$.
EM/non-EM stress ratio mean	4.3828%	Frozen representative alpha ancestry sample.
EM/non-EM stress ratio max	11.1860%	Largest sampled point.
R_reference, Z_reference	0.6680425, 0.75	Numerical chart scaffolding; not promoted observables.

Appendix B. Internal Provenance Map

V13.12 is written in structural order. The following internal artifacts preserve the chronological evidence. They are not all intended as public standalone papers; they are the audit trail behind the presentation.

Ref	Artifact	Role in V13.12
[P1] C17G15.1	Good-Enough Isolated Electron V1 Structure Freeze	Freezes the practical V1 static state at a validated Pareto floor without new source physics.
[P2] C17G16	Final Stability Audit	Floquet, positive-invariant, long-cycle, hostile-profile, instability-boundary, and free-geodesic controls.
[P3] C17G17	Frozen-State Physical Readout Ledger	Clock, rest momentum, Pauli identity, source normalization, stationary radiation scope, and local closure-gravity readouts from the exact frozen SHA.
[P4] C17G18	One-State Consistency / No-Retuning Audit	Exposes alpha ancestry, retypes R/Z scaffolding, verifies one-state accounting, and clears the final freeze blocker.

Ref	Artifact	Role in V13.12
[P5] C17G19	Final Claim-Boundary and Reproduction Freeze	Seals the parameter/interface ledger and full maintained Electron V1 freeze.
[P6] C17G9	Strict Gordon/Pauli Hot-Swap	Establishes the native $L=0$ rest branch, one-count Hilbert/Belinfante firewall, and charged continuation at reduced scope.
[P7] C17G10	Curved Sourced-Maxwell/Gordon Action	Solves curved stationary Maxwell response and folds it into the profiled charged common-action continuation.
[P8] C17G11	Full 50D Charged Event Audit	Lifts charged continuation into the full recovered material/metric Jacobian space and localizes remaining stationarity debt.
[P9] C17G14 live R/Z gate	Route coordinate ancestry audit	Shows R/Z are recognized but give only microscopic incremental improvement; values are retained as scaffolding, not observables.
[P10] SFT Alpha-137 standalone note	Companion coupling program	Earlier independent program leaves numerical alpha open and targets a microscopic projection/stiffness mechanism. Companion context only.

Appendix C. Reproduction and Freeze Rule

Canonical state: SFT_ELECTRON_V1_STRUCTURE_GOOD_ENOUGH_FROZEN.npz. The SHA-256 hash above is the restart anchor. Any future calculation claiming continuity with V13.12 must either reproduce that state or explicitly declare a child/fork that changes it.

- Formation work must treat the frozen maintained state as a target/output class rather than silently using its downstream clock to select formation.
- Alpha/coupling work may replace the external EM normalization only in a declared interface child program; the current 1/137 ancestry must never be retroactively described as a prediction.
- Exterior gravity or electromagnetism must add explicit asymptotic matching without retuning the internal state merely to hit a far-field observable.
- Interaction work must solve joint states. It must not model a real interaction as an invisible external force acting on a permanently isolated electron.
- If a downstream theorem genuinely requires a changed maintained state, fork a new Electron version rather than rewriting V13.12.

Public-document note: this v0.2 file is a presentation draft built on the frozen physics ledger. Editorial changes, figure redesign, bibliography expansion, and explanatory restructuring do not reopen the frozen numerical state.

END OF SFT V13.12 PUBLIC DRAFT v0.2